

Data Storytelling & Statistical Validation

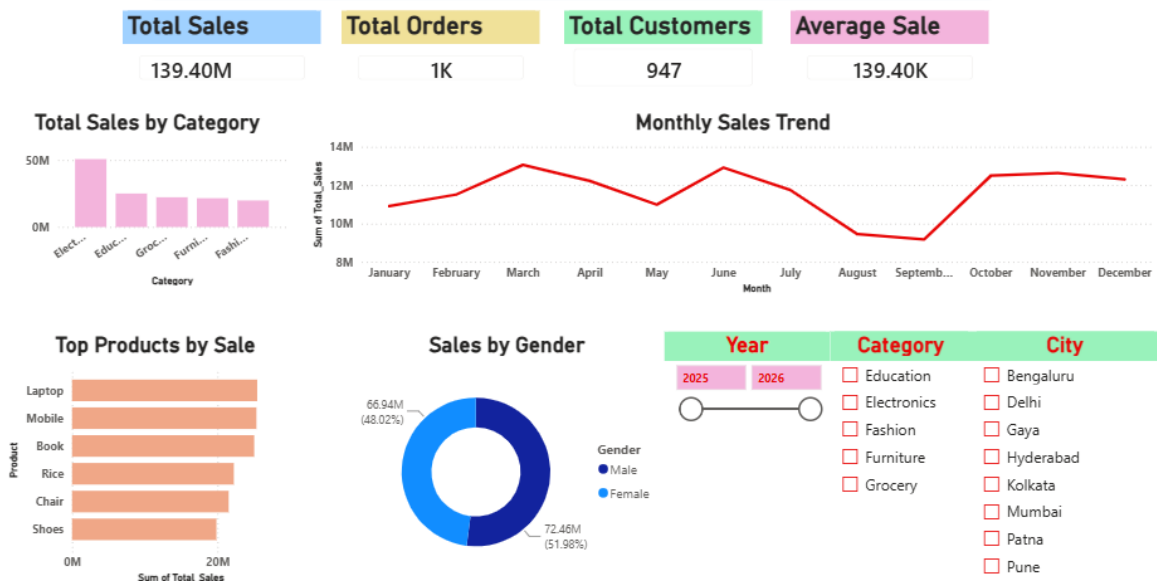
Business Story

The analysis begins with an overview of the company's overall performance. During the selected period, the business generated **₹139.40 million in total sales** from approximately **1,000 orders**, serving **947 unique customers**. The **average sale value of ₹139.40K** indicates that customers are making relatively high-value purchases.

These key performance indicators suggest that the business has a strong customer base and consistent sales activity. While the number of customers is slightly lower than the number of orders, it indicates that some customers have made repeat purchases, which is a positive sign of customer retention.

From a business perspective, maintaining customer satisfaction while attracting new customers should remain a priority. Loyalty programs, personalized offers, and excellent customer service can further improve repeat purchases and overall revenue.

Deep-Dive Analysis & Interactive Dashboard



1. KPI Cards (Overall Business Performance)

Business Story

- Total Sales reached **₹139.40M**.
- A total of **1,000 orders** were placed.
- The business served **947 customers**.
- Average sale per order was **₹139.40K**.

Business Insight

- Strong sales and a large customer base indicate healthy business performance.

Recommendation

- Focus on retaining customers and increasing repeat purchases.

2. Total Sales by Category

Business Story

- **Electronics** generated the highest sales.
- Education ranked second.
- Fashion contributed the lowest sales.

Business Insight

- Electronics is the major revenue-driving category.

Recommendation

- Increase investment and promotions for Electronics.
 - Improve marketing strategies for low-performing categories.
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3. Monthly Sales Trend

Business Story

- Sales remained stable throughout the year.
- Highest sales were recorded in **March, June, October, and November**.
- Sales dropped during **August and September**.

Business Insight

- Sales are influenced by seasonal demand.

Recommendation

- Run promotional campaigns during low-sales months.
 - Prepare inventory before peak sales periods.
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4. Top Products by Sale

Business Story

- **Laptop** is the best-selling product.
- Mobile and Books also generated strong sales.
- Shoes recorded the lowest sales among the top products.

Business Insight

- High-value products contribute significantly to overall revenue.

Recommendation

- Promote top-selling products through bundles and discounts.
 - Improve visibility of lower-selling products.
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5. Sales by Gender

Business Story

- Male customers contributed **51.98%** of total sales.
- Female customers contributed **48.02%** of total sales.
- Sales contribution is almost balanced.

Business Insight

- Both customer groups contribute nearly equally to revenue.

Recommendation

- Continue balanced marketing strategies for both genders.
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6. Dashboard Summary

Key Findings

- Electronics is the highest-performing category.
- Laptop is the top-selling product.
- Sales peak in **March, June, October, and November**.
- Male and female customers contribute almost equally.
- Overall business performance is strong with **₹139.40M** in sales.

Final Recommendation

- Expand the Electronics category.
 - Focus marketing during low-sales months.
 - Retain existing customers through loyalty programs.
 - Use dashboard insights for data-driven business decision
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7. Hypothesis Testing

Business Question

Is there a statistically significant difference in the average sales between male and female customers?

Null Hypothesis (H_0)

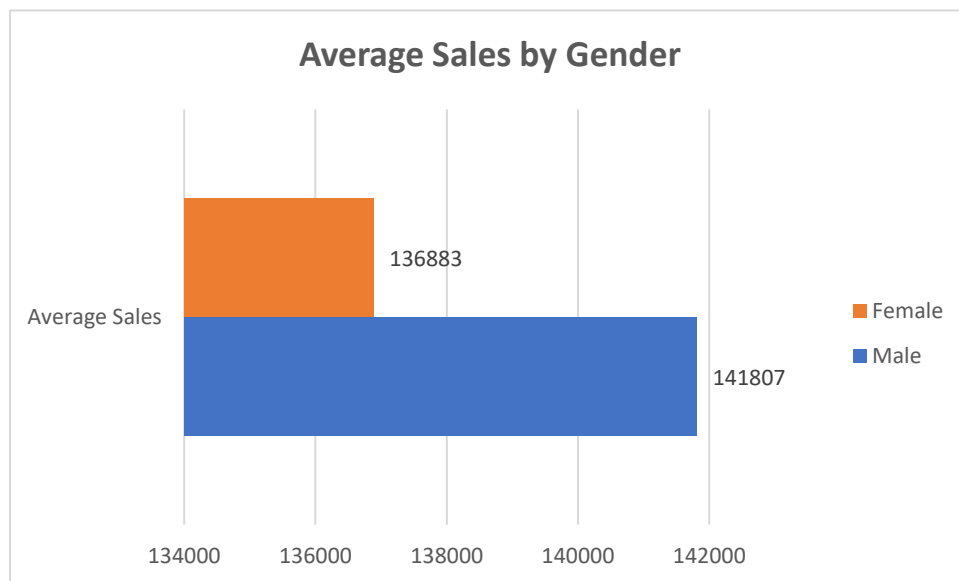
There is no statistically significant difference in the average sales between male and female customers.

Alternative Hypothesis (H_1)

There is a statistically significant difference in the average sales between male and female customers.

Results

- Male Average Sales: ₹141807.3397847358
- Female Average Sales: ₹ 136883.20862985688
- T-Statistic: 0.6820125406168254
- P-Value: 0.49538926844824827



8. Interpretation

The p-value is greater than the significance level of 0.05. Therefore, we fail to reject the null hypothesis. This indicates that the observed difference in average sales between male and female customers is not statistically significant and is likely due to random variation rather than a true difference.

9. Business Conclusion

Although male customers contribute slightly more sales than female customers, the difference is not statistically significant. Therefore, the company does not need separate marketing strategies based solely on gender. Instead, it should continue focusing on high-performing product categories, customer retention, and seasonal marketing campaigns.